Stadistical Physics

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Índice general

Capítulo 1

Classical atomic gas

1.1. Ideal systems

In an ideal system I can express the system's hamiltonian as

$$H = \sum_i h_i \tag{1.1}$$

Applying this in the canonical ensemble framework's probability we have

$$p(C) = \frac{e^{\beta H(C)}}{Z} = \frac{e^{\beta \sum_i h_i}}{Z} = \prod_i \frac{e^{\beta h_i}}{Z} \tag{1.2}$$

Nota

The previous result is consistence with the stadistical result: if $A \cap B = 0$ then $P(A \cup B) = P(A)P(B)$

1.1.1. Examples

If we have a potential $V(r_{ij})$, this is a potential that depends on the relative position of the particle i with particles j. I cannot separate the hamiltonian because the particle i dependes on surrounding particles.

$$V(r_{ij}) \implies H \neq \sum_i h_i \tag{1.3}$$

If we have a potential $V(r_i)$, this is a potential that only depends on the particle's position. Then I can separate the hamiltonian.

$$V(r_i) \implies H = \sum_i h_i = \sum_i \frac{p_i^2}{2m_i} + \sum_i V(r_i) \tag{1.4}$$

The Ising Model that we have seen in Computational Physics has the following hamiltonian

$$H = -\frac{J}{2} \sum_{ij}^{N} s_{ij} (s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \tag{1.5}$$

As we can see in the equation, the hamiltonian of the particle i-j depend on the surrounding, then we cannot express the hamiltonian as sums of each particles' hamiltonians.

$$H = -\frac{J}{2} \sum_{ij}^{N} s_{ij} (s_{i+1,j} + s_{i-1,j} + s_{i,j+1} + s_{i,j-1}) \neq \sum_{i} h_i \tag{1.6}$$

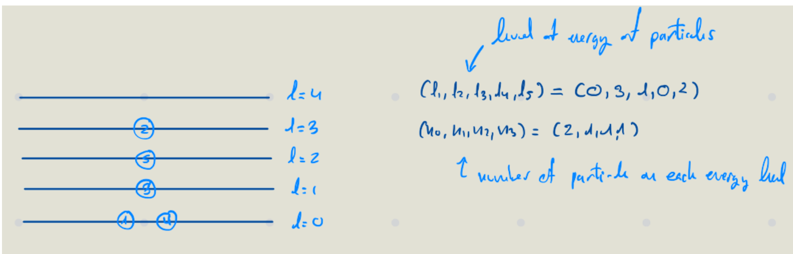
If we have the following hamiltonian, we can separate it in sum of particles' hamiltonian with $h_i = -J s_i$.

$$H = -J \sum_i s_i = \sum_i h_i = \sum_i -J s_i \tag{1.7}$$

1.2. Boltzmann Gas

The Boltzmann Gas it is the common limit of fermions and bosons.

1.2.1. In canonical ensemble



$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta \sum_i n_i \epsilon_i} \tag{1.8}$$

Due to the condition $\sum_i n_i = N$, we cannot calculate the partition function as the sum of partition functions. But we can express the total energy as sum of energy associated to level l.

$$E = \sum_i n_i \epsilon_i = \sum_i E_{l_i} \tag{1.9}$$

Applying this in the partition function we obtain

$$Z = \sum_{l_1,\dots,l_N} e^{\beta \sum_i \epsilon_{l_i}} \tag{1.10}$$

We need take account the degeneracy. As the particle are **indistiguishable**, we can consider all the particle as the same, then we only nned to know how to distribuite N same particles in l levels.

A set of $\{n_l\}$ correspond $\frac{N!}{\prod_i n_l!}$ possible values of $\{l_1, \dots, l_N\}$.

$$Z = \sum_{\{n_i\}, \sum_i n_i = N} e^{-\beta E} = \sum_{\{n_i\}, \sum_i n_i = N} \frac{\prod_i n_l!}{N!} e^{-\beta \sum_i n_i \epsilon_i} \tag{1.11}$$

1.2.2. Boltzmann approximation

The Boltzmann approximation is $n_l! \approx 1$. This is when $\rho \Lambda_T^3 \ll 1$ with $\Lambda_T^3 = h/\sqrt{2\pi m K T}$. This approximation is good when $T \upuparrows$ or/and $\rho \downdownarrows$ (very diluete cases). Under this condition, bosons and fermions **behave the same way**.

With the Boltzmann approximation, the partition function is

$$Z = \sum_i \frac{1}{N!} e^{-\beta \sum_i \epsilon_{l_i}} = \frac{Z_1^N}{N!} \tag{1.12}$$

1.2.3. Particles in a 3D potential well

Let's calculate the partition function in this system. Let's assume that

$$Z = \sum \frac{1}{N!} (Z_1)^N \tag{1.13}$$

Then let's calculate Z_1

A single particle's energy has two contribution, one due to the traslational degrees of freedom and one due to the internal degrees of freedom.

$$\epsilon = \epsilon_{tr} + \epsilon_{in} \tag{1.14}$$

Then the particle function of a single particle is

$$Z_1 = \sum_{l_{tr}} \sum_{l_{in}} e^{-\beta\epsilon_{tr}-\beta\epsilon_{in}} = Z_1^{tr} Z_1^{in} \tag{1.15}$$

In RAW we developed the solution of Schrödinger's Equation in a potential well in 1D. From this solution we obtained the expression for energy, which is related to the momentum.

$$\epsilon = \frac{nh}{4Lm} \tag{1.16}$$

Using this for the traslational energy expression

$$\epsilon_{tr} = \frac{p_x^2 + p_y^2 + p_z^2}{2m} = \frac{\Lambda}{2m}\left(\frac{n_x^2 h^2}{4L^2} + \frac{n_y^2 h^2}{4L^2} + \frac{n_z^2 h^2}{4L^2}\right) \tag{1.17}$$

Let's calculate the partition function

$$Z_1^{tr} = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2}(n_x^2+n_y^2+n_z^2)} \tag{1.18}$$

Let's take a look how to solve individualy

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} \tag{1.19}$$

If we notice, p is related to the length of the well: $p = hn/2L$, then $\Delta p = p_{n+1} - p_n = h/2L$. If we consider a very large L, this is $L \to \infty$ then $\Delta p = dp$, let's use this to simplify the summatory. Let's change of variable: $n = pL/h\pi$

$$\sum_{n=1}^{\infty} e^{-\beta \frac{h^2}{8mL^2} n^2} = \sum_{p=\frac{h}{2L}}^{\infty} e^{-\beta \frac{p^2}{2m}} \tag{1.20}$$

We introduce Δp in the summatory and we are going to perform a limit, then we can covert the summatory into an integral.

$$\underset{\to\infty}{lim} \sum_{p=\frac{h}{2L}}^{\infty} \frac{2L}{h} \Delta p \; e^{-\beta \frac{p^2}{2m}} = \int_0^{\infty} e^{-\beta \frac{p^2}{2m}} \; dp = \frac{1}{2} \int_{-\infty}^{\infty} e^{-\beta \frac{p^2}{2m}} \; dp \tag{1.21}$$

Substituing the partition function with this integral we have

$$\frac{1}{h^3}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} dp_x\, dp_y\, dp_z\, e^{-\frac{\beta}{2m}(p_x^2+p_y^2+p_z^2)} \tag{1.22}$$

Taking into account the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}}\, dx = \sqrt{2\pi\sigma^2} \tag{1.23}$$

It follows that:

$$Z_1^{(tr)} = \frac{L^3}{h^3}(2\pi mkT)^{3/2} = \frac{V}{\Lambda_T^3} \tag{1.24}$$

Then the partition funcion is

$$Z = \frac{1}{N!}\left(\frac{V}{\Lambda_T^3}\right)^N \tag{1.25}$$

The previous result coincide with the classical ideal monoatomic gas.

1.2.4. Thermodynamic properties of the BG

$$A(T,V,N) = -kT\ln Z_{BG} = -NkT\ln z_1^{(tr)} + \underbrace{kT\ln N!}_{NkT\ln N - NkT} - NkT\ln z_1^{(in)}$$

Using the *Stirling approximation* and knowing that $z_1^{(tr)} = \frac{V}{\Lambda_T^3}$:

$$A(T,V,N) = -NkT + NkT\ln\left(\tfrac{V}{N}\right) - NkT\ln\Lambda_T^3 + NkT\ln z_1^{(in)}$$

Defining the free energy per particle $a(T,n) = \frac{A(T,V,N)}{N} = -kT + kT\ln(n\Lambda_T^3) + a_{in}(T)$, where $a_{in}(T) = -kT\ln z_1^{(in)}$ and $n = \frac{N}{V}$.

Considering $dA = -SdT - PdV + \mu dN$:

Eq. of State (Pressure)

$$P = -\left(\frac{\partial A}{\partial V}\right)_{T,N} \tag{1.26}$$

Since $-kT\ln z_1^{(in)}$ has no V dependence, only the translational degrees of freedom contribute to the equation of state (pressure):

$$P = -N\left(-\frac{\partial}{\partial V}(kT\ln(\frac{V}{N}))\right) = \frac{NkT}{V} \implies \text{Boltzmann gas behaves as an ideal gas} \tag{1.27}$$

The internal partition function $z^{(in)}$ contributes to the entropy, free energy, and specific heat.

Entropy

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,N} \implies s = -\left(\frac{\partial a}{\partial T}\right)_n = -\left[k\ln(n\Lambda_T^3) - k + a'_{in}(T) + \frac{kT\cdot 3\Lambda_T^2\cdot\Lambda'_T(T)}{\Lambda_T^3}\right] \tag{1.28}$$

Given that $\Lambda_T = \frac{h}{\sqrt{2\pi mkT}} \implies \Lambda'_T(T) = -\frac{\Lambda_T(T)}{2T}$, we derive the **Sackur-Tetrode equation**:

$$s(T,n) = k\left[\frac{5}{2} - \ln(n\Lambda_T^3)\right] - a'_{in}(T) \tag{1.29}$$

It appears that $s(T\to 0, n) \neq 0$. This occurs because we are using the BG model; therefore, T must be high and/or n must be low for the approximation to hold.

Internal Energy

$$A = U - TS \implies U = A + TS \implies u = a + Ts \tag{1.30}$$

$$u = a + Ts = \frac{3kT}{2} + a_{in}(T) - Ta'_{in}(T) \tag{1.31}$$

The $\frac{3kT}{2}$ term is the contribution from the equipartition theorem. We have a contribution of $\frac{1}{2}kT$ for each quadratic term in the Hamiltonian $H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + \dots$

Specific Heat

$$c_v = \left(\frac{\partial u}{\partial T}\right)_n = \frac{3}{2}k - Ta''_{in}(T) \tag{1.32}$$

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1.2.5. Role played by internal degrees of freedom

$A_{in} = -NkT\ln z_1^{(in)}$ (Free energy)

$S_{in} = -\left(\frac{\partial A_{in}}{\partial T}\right)_{V,N} = Nk\left(\ln z_1^{(in)} + T\frac{\partial}{\partial T}\ln z_1^{(in)}\right)$ (Entropy)

$U_{in} = A_{in}+TS_{in} = -NkT\ln z_1^{(in)}+NkT\left(\ln z_1^{(in)} + T\frac{\partial}{\partial T}\ln z_1^{(in)}\right) = NkT^2\frac{\partial}{\partial T}\ln z_1^{(in)}$ (Internal ene

$C_v^{in} = \frac{\partial}{\partial T}\left[NkT^2\frac{\partial}{\partial T}\ln z_1^{(in)}\right]_{V,N}$ (Specific heat)

If $z_1^{(in)} = z_{rot}\cdot z_{vib}\cdot z_{elec}$:

$$\begin{aligned}
A^{(in)} &= -kT\ln z^{(in)} = A_{rot} + A_{vib} + A_{elec} \\
S^{(in)} &= -\left(\frac{\partial A^{(in)}}{\partial T}\right) = S_{rot} + S_{vib} + S_{elec} \\
U^{(in)} &= U_{rot} + U_{vib} + U_{elec} \\
C_v^{(in)} &= C_v^{rot} + C_v^{vib} + C_v^{elec}
\end{aligned}$$

Activation of internal energy d.o.f.

$z_1^{(in)} = \sum_n e^{-\beta\varepsilon_n^{(in)}}$ where n is the quantum number.

Spectrum of the internal d.o.f. for the internal d.o.f.

- $\Delta\varepsilon > \varepsilon_1 - \varepsilon_0$ (gap or activation energy)
- $T_a = \frac{\Delta\varepsilon}{k}$ (activation temperature)

First limit case: Thermal energy (kT) much smaller than $\Delta\varepsilon$

$kT \ll \Delta\varepsilon \implies \beta\Delta\varepsilon \gg 1$. Taking $\varepsilon_0, \varepsilon_1, \varepsilon_2\ldots$ and assuming $\ln e^{-\beta\varepsilon_0} = -\beta\varepsilon_0 = -\frac{\varepsilon_0}{kT}$:

$$z^{in} = \sum_\varepsilon g(\varepsilon)e^{-\beta\varepsilon} \approx g(\varepsilon_0)e^{-\beta\varepsilon_0}\left(1 + \frac{g(\varepsilon_1)}{g(\varepsilon_0)}e^{-\beta\Delta\varepsilon} + \ldots\right) \approx e^{-\beta\varepsilon_0}g(\varepsilon_0) \qquad (1.33)$$

Then $C_v^{in} = k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}\ln z^{in}\right] \approx k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}\left(-\frac{\varepsilon_0}{kT}\right)\right] = k\frac{\partial}{\partial T}\left[T^2\frac{\varepsilon_0}{kT^2}\right] = 0. \implies$ In this case, internal degrees of freedom do not contribute to C_v.

Second limit case: Thermal energy (kT) much larger than $\Delta\varepsilon$

$kT \gg \Delta\varepsilon$: We can take the continuous limit of the specific energies and the system behaves as classical. $\implies$ Equipartition theorem holds $\implies U = \alpha NkT$ ($C_v =$ constant).

Graph of $C_v = C_v^{(tr)} + C_v^{(in)}$

- At low T: $C_v^{in} = 0$, so $C_v = \frac{3}{2}k$.

- At high T ($kT \gg \Delta\varepsilon$): C_v increases and reaches a new plateau due to the activation of internal degrees of freedom.

To study the vibrational d.o.f., we are going to study diatomic molecules.

$V(r)$ Taylor expansion around r_0, the equilibrium position:

$$V(r) \approx V(r_0) + \cancel{V'(r_0)(r-r_0)} + \frac{1}{2}V''(r_0)(r-r_0)^2 + \cdots \implies \frac{1}{2}kx^2 \to \text{Harmonic Oscillator} \tag{1.34}$$

Now we consider Schrödinger eq:

$$\hat{H}\psi = E\psi \to -\frac{\hbar^2}{2\mu}\frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi \to \omega = \sqrt{\frac{k}{\mu}} \to \frac{d^2\psi}{dx^2} + \frac{2\mu}{\hbar^2}E = \dots \tag{1.35}$$

The solution to this is: $\varepsilon_n = \hbar\omega(n+1/2)$ with $\omega = \sqrt{\frac{k}{\mu}}$, where $n = 0, 1, 2, 3\dots$

Partition function:

$$z_1^{vib} = \sum_{n=0}^{\infty} e^{-\beta\varepsilon_n} = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} = e^{-\frac{\beta\hbar\omega}{2}} \sum_{n=0}^{\infty} (e^{-\beta\hbar\omega})^n = e^{-\frac{\beta\hbar\omega}{2}} \frac{1}{1-e^{-\beta\hbar\omega}} = \frac{e^{-\frac{\beta\hbar\omega}{2}}}{1-e^{-\beta\hbar\omega}} \cdot \frac{e^{\frac{\beta\hbar\omega}{2}}}{e^{\frac{\beta\hbar\omega}{2}}} = \frac{}{e^{\frac{\beta\hbar\omega}{2}} -} \tag{1.36}$$

Using the identity $2\sinh(x) = e^x - e^{-x}$:

$$z_1^{vib} = [2\sinh(\beta\hbar\omega/2)]^{-1} \implies Z^{vib} = (z_1^{vib})^N = [2\sinh(\beta\hbar\omega/2)]^{-N} \tag{1.37}$$

Free energy: $A_{vib} = -kT\ln Z = N[\frac{1}{2}\hbar\omega + kT\ln(1-e^{-\beta\hbar\omega})] \implies$ does not depend on V.

Therm. pressure: $P = -(\frac{\partial A}{\partial V})_{T,N} = 0 \implies$ vibrational d.o.f. do not contribute to the pressure.

Entropy: $s_{vib} = -(\frac{\partial a}{\partial T})_n = Nk\left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega}-1} - \ln(1-e^{-\beta\hbar\omega})\right]$

Internal energy: $u_{vib} = a + Ts = Nk\left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega}-1}\right]$

Specific heat: $c_{v,vib} = \frac{1}{N}(\frac{\partial U}{\partial T})_{V,N} = k(\beta\hbar\omega)^2\frac{e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2}$

As we see, the z^{vib} which is an internal degree of freedom does not contribute to the state of stage equation (pressure).

$\Theta_v \to$ *vibrational activation temperature* Activation energy: $\varepsilon_n = \hbar\omega(n+1/2)$ $k\Theta_v = \hbar\omega \implies \Theta_v = \frac{\hbar\omega}{k} \implies c_{v,vib} = k(\frac{\Theta_v}{T})^2\frac{e^{\Theta_v/T}}{(e^{\Theta_v/T}-1)^2}$

We shall distinguish two cases:

- $T \gg \Theta_v \implies \frac{\Theta_v}{T} \ll 1$: $c_{v,vib} \approx k(\frac{\Theta_v}{T})^2 \frac{1+\dots}{(1+\frac{\Theta_v}{T}+\dots-1)^2} = k\frac{(\Theta_v/T)^2}{(\Theta_v/T)^2} = k$
- $T \ll \Theta_v \implies \frac{\Theta_v}{T} \gg 1$: $c_{v,vib} \approx k(\frac{\Theta_v}{T})^2 \frac{e^{\Theta_v/T}}{(e^{\Theta_v/T})^2} \xrightarrow{T\to 0} 0$

Rotational d.o.f.

Rigid rotor model: $T = \frac{1}{2}I\omega^2$ (rotational kinetic energy) with $I = \mu r_0^2$ and $\mu = \frac{m_1 m_2}{m_1+m_2}$. $L = I\omega \implies \vec{L} = \vec{r}\times\vec{p}$, then $T = \frac{L^2}{2I}$.

$\hat{H}\psi = E\psi \to \frac{L^2}{2I}\psi = E\psi$ $\vec{L} = \vec{r}\times\vec{p} = \dots \implies L^2 = L_x^2 + L_y^2 + L_z^2$

Using spherical coordinates: $x = r\sin\theta\cos\phi$ $y = r\sin\theta\sin\phi$ $z = r\cos\theta$

$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$

The solution of this Schrödinger equation is: $\psi(\theta,\phi) = AY_{lm}(\theta,\phi) = Y_{lm}(\theta,\phi)$ where $l = 0,1,2\dots$ and $m = -l,\dots,l$.

The eigenvalue is: $\varepsilon_l = \frac{\hbar^2}{2I}l(l+1)$ $\quad H\psi = E\psi$

$$z_1 = \sum_{l=0}^{\infty}(2l+1)e^{-\beta\varepsilon_l} = \sum_{l=0}^{\infty}(2l+1)e^{-\beta\frac{\hbar^2}{2I}l(l+1)} \tag{1.38}$$

For each l, there are $(2l+1)$ possible values of m_l that give us the same energy. I'm taking into account the degeneracy.

Temperature: $\Theta_{rot} = \frac{\Delta\varepsilon}{k} = \frac{\hbar^2}{2Ik}$ $\implies \Theta_{rot} = \frac{\hbar^2}{2Ik}$ (added by convenience). $\Delta\varepsilon = \varepsilon_1 - \varepsilon_0 = \frac{\hbar^2}{2I}\cdot 1\cdot 2 - 0 = \frac{\hbar^2}{I}$

Then $z_1^{rot} = \sum_{l=0}^{\infty}(2l+1)e^{-\frac{\Theta_{rot}}{T}l(l+1)}$. This summation is very difficult to compute, so we are going to perform approximations.

A particular case: $T \ll \Theta_{rot} \implies \Theta_{rot}/T \gg 1$

$\implies z_1^{rot} \approx 1 + 3e^{-2\Theta_{rot}/T} + 5e^{-6\Theta_{rot}/T} + \dots$ (the rest is negligible).
Using $\ln(1+x) \approx x$ with $x \ll 1$:

$$c_{v,rot} = k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}\ln z_1^{rot}\right] \approx k\frac{\partial}{\partial T}\left[T^2\frac{\partial}{\partial T}3e^{-2\Theta_{rot}/T}\right] = 3k\left(\frac{2\Theta_{rot}}{T}\right)^2 e^{-2\Theta_{rot}/T} \tag{1.39}$$

When $\frac{T}{\Theta_{rot}} \to 0 \implies c_{v,rot} \to 0$. We are in this regime (the curve starts when the degree of freedom is activated).

We approximate the sum ($\sum f(n)\dots$) with the **Euler-Maclaurin Formula**:

$$\sum_{n=a}^{b} f(n) \approx \int_a^b f(x)dx + \frac{f(a)+f(b)}{2} + \sum_{j=1}^{p}\frac{B_{2j}}{(2j)!}[f^{(2j-1)}(b) - f^{(2j-1)}(a)] + \dots \tag{1.40}$$

where B_{2j} are Bernoulli numbers and $f^{(n)}$ are derivatives.

Then: $z_1^{rot} = \sum_{l=0}^{\infty}(2l+1)e^{-\frac{\Theta_{rot}}{T}l(l+1)}$ Taking $f(x) = (2x+1)e^{-\frac{\Theta_{rot}}{T}x(x+1)}$:

- $\int_0^\infty f(x)dx = \frac{T}{\Theta_{rot}}$
- $f(0) = 1, \quad f(\infty) = 0$
- $f'(0) = \frac{\Theta_{rot}}{T}(1 - 2 \cdot 0) = \frac{\Theta_{rot}}{T}, \quad f'(\infty) = 0$

$z_1^{rot} \approx \frac{T}{\Theta_{rot}} + \frac{1}{2} - \frac{1}{12}\left[\frac{\Theta_{rot}}{T}\right] + \frac{1}{720}f'''(0)\dots$

$$z_1^{rot} \approx \frac{T}{\Theta_{rot}} + \frac{1}{3} + \frac{1}{15}\frac{\Theta_{rot}}{T} + \frac{4}{315}\left(\frac{\Theta_{rot}}{T}\right)^2 \dots \to \text{let's study the another limit case} \tag{1.41}$$

When $T \gg \Theta_{rot} \implies \frac{T}{\Theta_{rot}} \gg 1$, then $z_1^{rot} \approx \frac{T}{\Theta_{rot}}$.

Then $c_{v,rot} = k\frac{\partial}{\partial T}\left[T^2 \frac{\partial}{\partial T}\ln\frac{T}{\Theta_{rot}}\right] = k.$

Combined Specific Heat Analysis

The graph shows the stages of activation for the degrees of freedom (C_v vs kT). In the rotational regime:

- There is a predicted maximum in the curve (indicated by the red circle), which can be measured experimentally.
- In this regime, as $T \gg \Theta_{rot}$:

$$c_{v,rot} \approx k\left[1 + \frac{1}{45}\left(\frac{\Theta_{rot}}{T}\right)^2 + \frac{16}{945}\left(\frac{\Theta_{rot}}{T}\right)^3 + \dots\right] \tag{1.42}$$

Activation Temperature Comparison

Molecule	$\mathbf{\Theta_{rot}}$ (K)	$\mathbf{\Theta_{vib}}$ (K)
H_2	86.4	6240
OH	15.2	4140
N_2	2.86	3340

For homonuclear molecules: $z_{homonuclear}^{rot} = \frac{z_{heteronuclear}^{rot}}{2}$

Note: As we see, $\Theta_{rot} \ll \Theta_{vib}$. It is much easier to rotate the molecule than to vibrate it. Therefore, at low temperatures, the molecule begins to rotate, and only at much higher temperatures (T) does it start vibrating.

Electronic d.o.f.

$$z^{elect} = \sum_n \Omega(E_n) e^{-\beta E_n} \tag{1.43}$$

where n is the quantum electronic level and Ω is the degeneracy.

In general, the activation temperature for electronic states T_{ae} is around $10^4 - 10^5$ K (very high temperatures). This is much larger than commonly reachable temperatures in standard lab conditions.

Since $kT \ll \Delta E = \frac{T_{ae}}{k}$:

$$Z^{elect} \approx \Omega(E_0) e^{-\beta E_0} \implies C_{v,elect} \approx 0 \tag{1.44}$$

We did this before!